

# List 12. Multinomial Logit

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## Contents

|          |                                |          |
|----------|--------------------------------|----------|
| <b>1</b> | <b>Regression fitting</b>      | <b>1</b> |
| <b>2</b> | <b>z-test</b>                  | <b>4</b> |
| 2.1      | Significance . . . . .         | 4        |
| <b>3</b> | <b>LR/Wald tests</b>           | <b>6</b> |
| 3.1      | Overall significance . . . . . | 6        |
| 3.2      | Joint significance . . . . .   | 7        |
| <b>4</b> | <b>Confidence intervals</b>    | <b>7</b> |
| <b>5</b> | <b>Prediction</b>              | <b>8</b> |

## 1 Regression fitting

#1. For the dataset `Diamond` consider a regression

`certification on price & carat.`

1. write down the specification
2. Evaluate matrices of regression design  $\mathbf{y}, \mathbf{X}$
3. Evaluate coefficients of the fitted model via regression specification (via formula)

4. Evaluate coefficients of the fitted model via matrices of regression design
5. Interpret coefficients of the fitted model

#2. For the dataset `Diamond` consider a regression

`colour on log(price) & carat.`

1. write down the specification
2. Evaluate matrices of regression design  $\mathbf{y}, \mathbf{X}$
3. Evaluate coefficients of the fitted model via regression specification (via formula)
4. Evaluate coefficients of the fitted model via matrices of regression design
5. Interpret coefficients of the fitted model

#3. For the dataset `Diamond` consider a regression

`clarity on price, carat, carat2.`

1. write down the specification
2. Evaluate matrices of regression design  $\mathbf{y}, \mathbf{X}$
3. Evaluate coefficients of the fitted model via regression specification (via formula)
4. Evaluate coefficients of the fitted model via matrices of regression design
5. Interpret coefficients of the fitted model

#4. For the dataset `diamonds` consider a regression

`cut on log(price), carat, x, y, z.`

1. write down the specification

2. Evaluate matrices of regression design  $\mathbf{y}, \mathbf{X}$
3. Evaluate coefficients of the fitted model via regression specification (via formula)
4. Evaluate coefficients of the fitted model via matrices of regression design
5. Interpret coefficients of the fitted model

#5. For the dataset `diamonds` consider a regression

`color on log(price), carat, carat2, x, y, z.`

1. write down the specification
2. Evaluate matrices of regression design  $\mathbf{y}, \mathbf{X}$
3. Evaluate coefficients of the fitted model via regression specification (via formula)
4. Evaluate coefficients of the fitted model via matrices of regression design
5. Interpret coefficients of the fitted model

#6. For the dataset `diamonds` consider a regression

`clarity on log(price), log2(price), carat, x, y, z.`

1. write down the specification
2. Evaluate matrices of regression design  $\mathbf{y}, \mathbf{X}$
3. Evaluate coefficients of the fitted model via regression specification (via formula)
4. Evaluate coefficients of the fitted model via matrices of regression design
5. Interpret coefficients of the fitted model

## 2 z-test

For each text calculate related critical values.

### 2.1 Significance

#1. For the dataset `Diamond` consider a regression

`certification on price & carat.`

1. Fit the regression
2. For each coefficient evaluate s.e., z-stat & p-value for z-test on the significance
3. Evaluate a related critical value
4. For each coefficient perform z-test for the significance and state a testing hypothesis. Make conclusions

#2. For the dataset `Diamond` consider a regression

`colour on log(price), carat, carat2.`

1. Fit the regression
2. For each coefficient evaluate s.e., z-stat & p-value for z-test on the significance
3. Evaluate a related critical value
4. For each coefficient perform z-test for the significance and state a testing hypothesis. Make conclusions

#3. For the dataset `Diamond` consider a regression

`clarity on log(price), log2(price), carat.`

1. Fit the regression
2. For each coefficient evaluate s.e., z-stat & p-value for z-test on the significance

3. Evaluate a related critical value
4. For each coefficient perform z-test for the significance and state a testing hypothesis. Make conclusions

#4. For the dataset **diamonds** consider a regression

cut on  $\log(\text{price})$ , carat, x, y, z.

1. Fit the regression
2. For each coefficient evaluate s.e., z-stat & p-value for z-test on the significance
3. Evaluate a related critical value
4. For each coefficient perform z-test for the significance and state a testing hypothesis. Make conclusions

#5. For the dataset **diamonds** consider a regression

color on  $\log(\text{price})$ , carat, carat<sup>2</sup>, x, y, z.

1. Fit the regression
2. For each coefficient evaluate s.e., z-stat & p-value for z-test on the significance
3. Evaluate a related critical value
4. For each coefficient perform z-test for the significance and state a testing hypothesis. Make conclusions

#6. For the dataset **diamonds** consider a regression

clarity on  $\log(\text{price})$ ,  $\log^2(\text{price})$ , carat, x, y, z.

1. Fit the regression
2. For each coefficient evaluate s.e., z-stat & p-value for z-test on the significance
3. Evaluate a related critical value

4. For each coefficient perform z-test for the significance and state a testing hypothesis. Make conclusions

#7. For the dataset `diamonds` consider a regression

`cut` on `log(price)`, `log2(price)`, `carat`, `carat2`.

1. Fit the regression
2. For each coefficient evaluate s.e., z-stat & p-value for z-test on the significance
3. Evaluate a related critical value
4. For each coefficient perform z-test for the significance and state a testing hypothesis. Make conclusions

### 3 LR/Wald tests

For each test calculate related critical values.

#### 3.1 Overall significance

#1. For the dataset `Diamond` consider a regression

`certification` on `price` & `carat`.

Test overall significance of the regression and state the testing hypothesis

#2. For the dataset `Diamond` consider a regression

`colour` on `price` & `carat`.

Test overall significance of the regression and state the testing hypothesis

#3. For the dataset `Diamond` consider a regression

`clarity` on `price` & `carat`.

Test overall significance of the regression and state the testing hypothesis

#4. For the dataset `diamonds` consider a regression

`cut` on `log(price)`, `carat`, `x`, `y`, `z`.

Test overall significance of the regression and state the testing hypothesis

#5. For the dataset `diamonds` consider a regression

`color on log(price), carat, carat2, x, y, z.`

Test overall significance of the regression and state the testing hypothesis

#6. For the dataset `diamonds` consider a regression

`clarity on log(price), log2(price), carat, x, y, z.`

Test overall significance of the regression and state the testing hypothesis

### 3.2 Joint significance

#7. For the dataset `diamonds` consider a regression

`cut on log(price), carat, x, y, z.`

Test joint significance of regressors `x`, `y`, `z` and state the testing hypothesis

#8. For the dataset `sleep75` consider a regression

`cut on log(price), carat, carat2, x, y, z.`

Test the significance of `carat` and state the testing hypothesis

#9. For the dataset `sleep75` consider a regression

`cut on log(price), log2(price), carat, x, y, z.`

Test the significance of `carat` and state the testing hypothesis

## 4 Confidence intervals

#1. For the dataset `Diamond` consider a regression

`certification on price & carat.`

Evaluate 90%-level confidence intervals for coefficients.

Which are significant at 10%-significant level?

#2. For the dataset `diamonds` consider a regression

cut on  $\log(\text{price})$ , carat, x, y, z.

Evaluate 95%-level confidence intervals for coefficients.

Which are significant at 5%-significant level?

#3. For the dataset `diamonds` consider a regression

color on  $\log(\text{price})$ , carat, x, y, z,  $x^2$ ,  $y^2$ ,  $z^2$ .

Evaluate 99%-level confidence intervals for coefficients.

Which are significant at 1%-significant level?

## 5 Prediction

#1. For the dataset `Diamond` consider a regression

certification on price & carat.

Consider diamonds with characteristics

|   | price | carat |
|---|-------|-------|
| 1 | 1260  | 0.3   |
| 2 | 1427  | 0.32  |
| 3 | 5193  | 0.71  |
| 4 | 9573  | 1.01  |

Predict the dependent variable for each individual in the sample and interpret predicted values. Round the answer to 3 decimal places.

#2. For the dataset `Diamond` consider a regression

colour on  $\log(\text{price})$ , carat,  $\text{carat}^2$ .

Consider diamonds with characteristics

|   | price | carat |
|---|-------|-------|
| 1 | 1555  | 0.31  |
| 2 | 6709  | 0.89  |
| 3 | 5030  | 0.73  |
| 4 | 9302  | 1.06  |

Predict the dependent variable for each individual in the sample and interpret predicted values. Round the answer to 3 decimal places.



#3. For the dataset Diamond consider a regression

clarity on  $\log(\text{price})$ ,  $\log^2(\text{price})$ , carat.

Consider diamonds with characteristics

|   | price | carat |
|---|-------|-------|
| 1 | 1126  | 0.31  |
| 2 | 1911  | 0.4   |
| 3 | 3651  | 0.57  |
| 4 | 9153  | 1.01  |

Predict the dependent variable for each individual in the sample and interpret predicted values. Round the answer to 3 decimal places.